Review: Identifying similar triangles

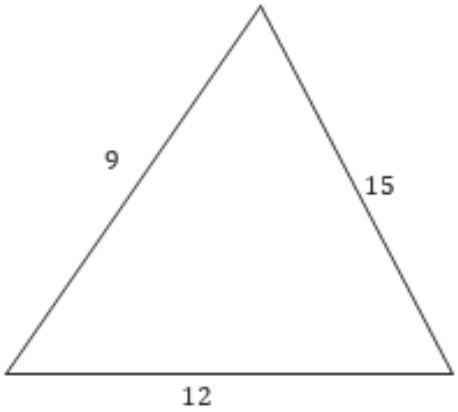
1 Consider the following sets of four triangles:

i Determine the similar triangles.

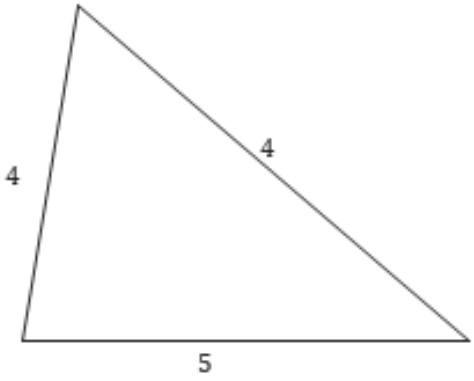
ii Give a suitable reason for their similarity.

a

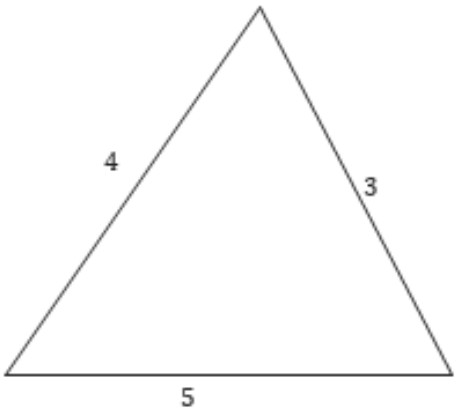
A



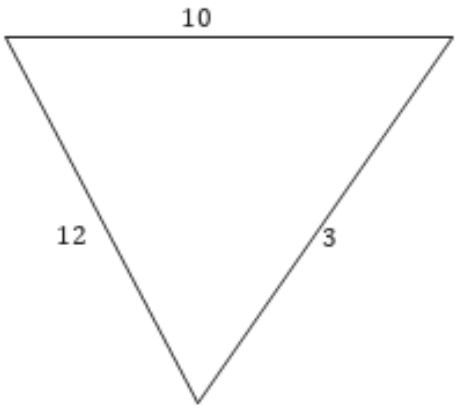
B



C



D

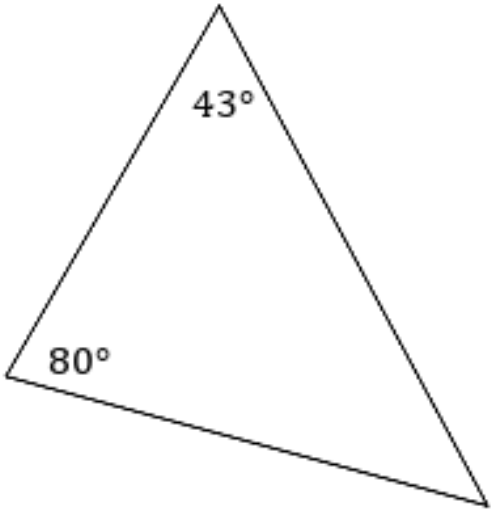


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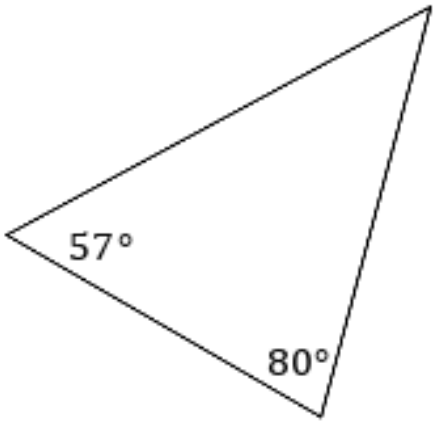
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2 Which of the following triangles are similar?

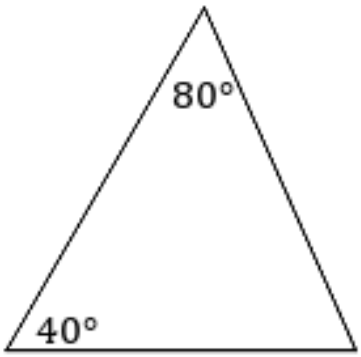
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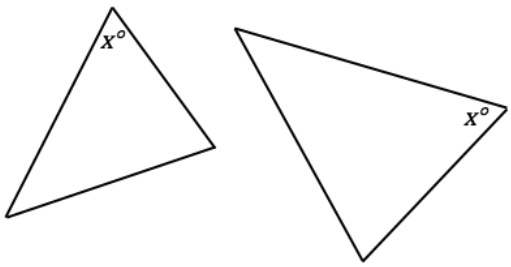
B



C

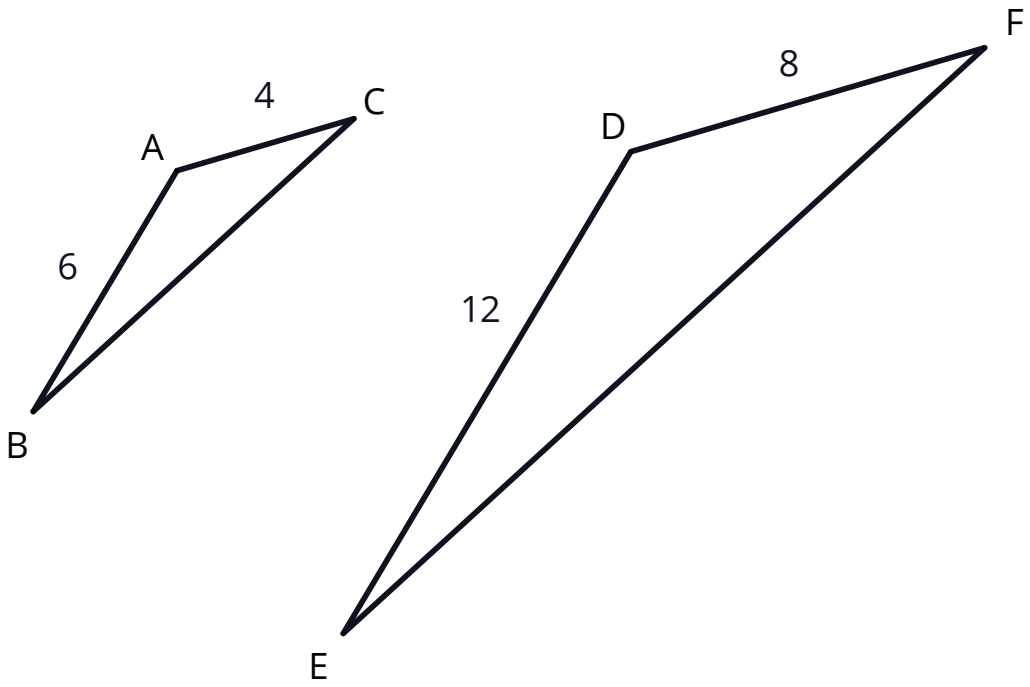


3 Consider the following triangles.
What is the minimum amount of extra information we would need to establish that these are similar triangles.



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4 Consider the following triangles.



Determine whether the following information, if added to the diagram, could be used to prove immediately that the triangles are similar.

a $\angle ABC \cong \angle DEF$

b $\angle BAC \cong \angle EDF$

c $\overline{BC} \cong \overline{EF}$

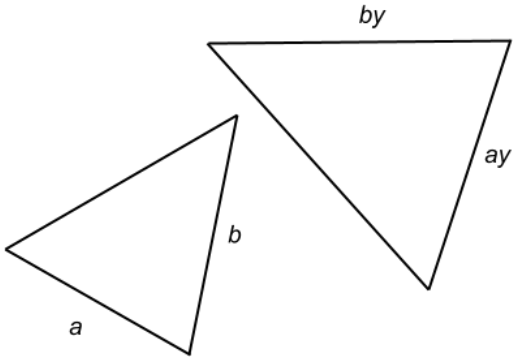
d $\angle ACB \cong \angle DFE$

5 For each of teh given pairs of triangles, determine whether there are enough information to show whether ot not the triangles are similar.

a



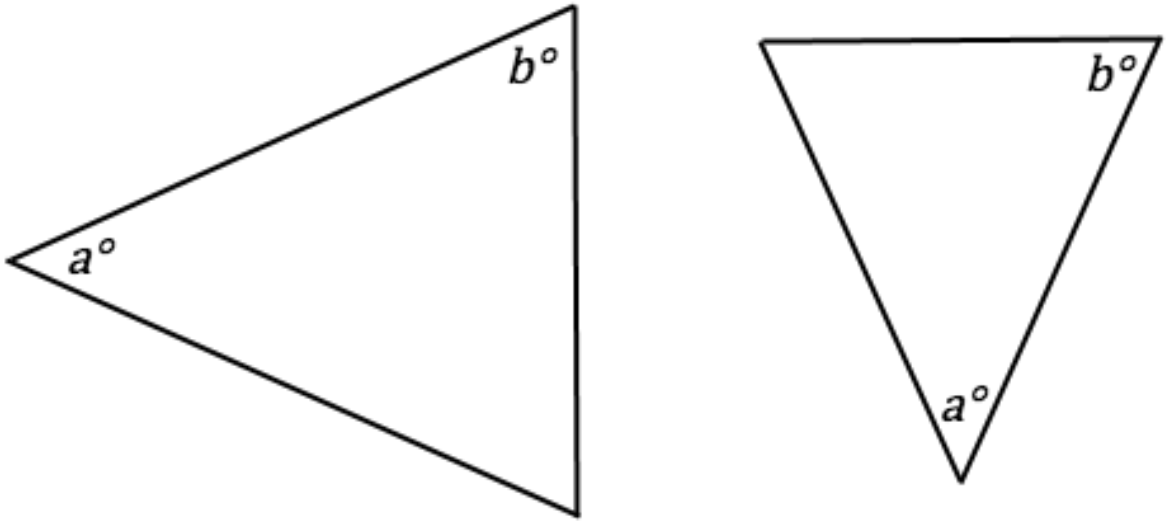
b



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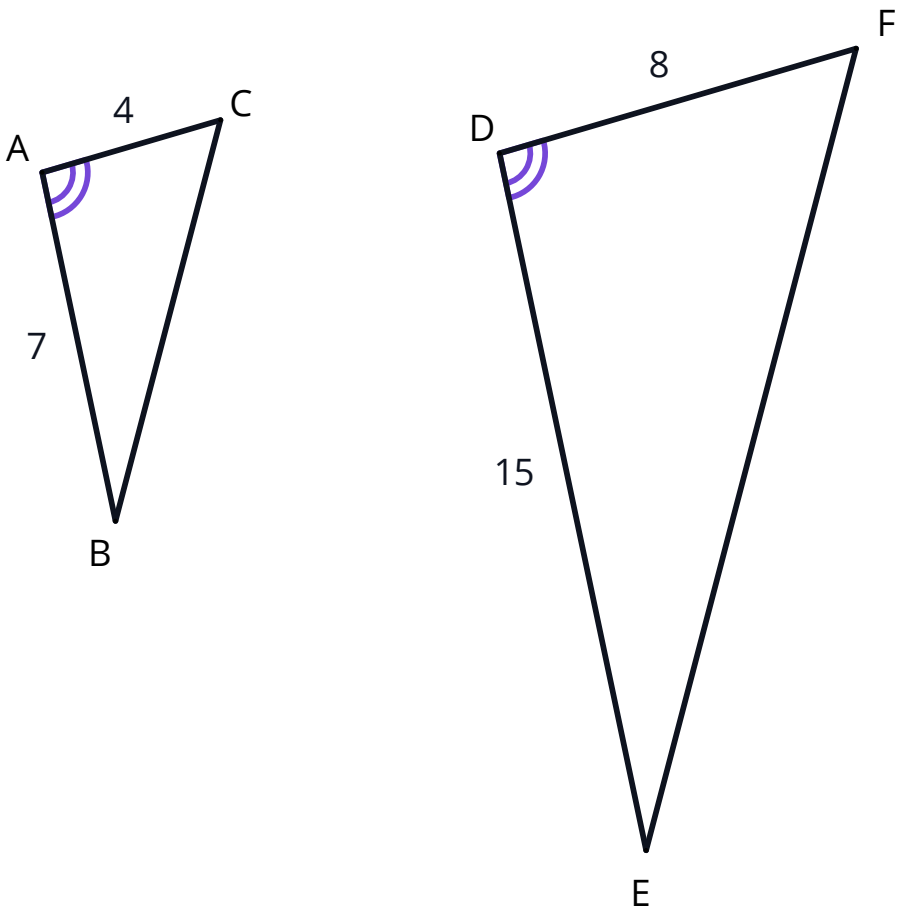
6 Determine whether the following pairs of triangles are similar or not. Give the reason for your answer.

a



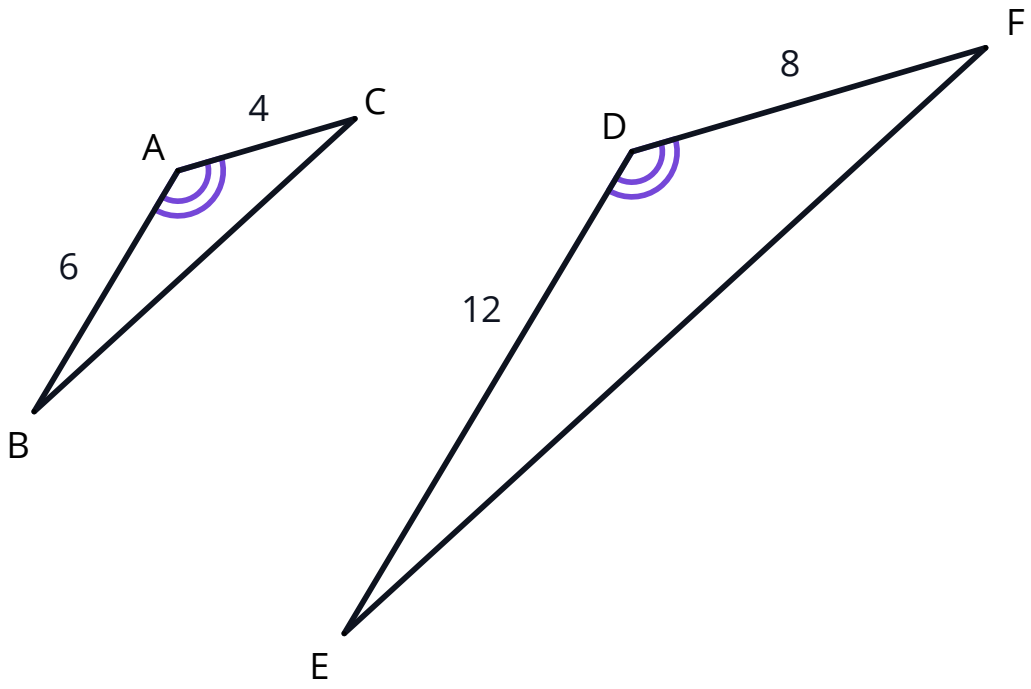
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b



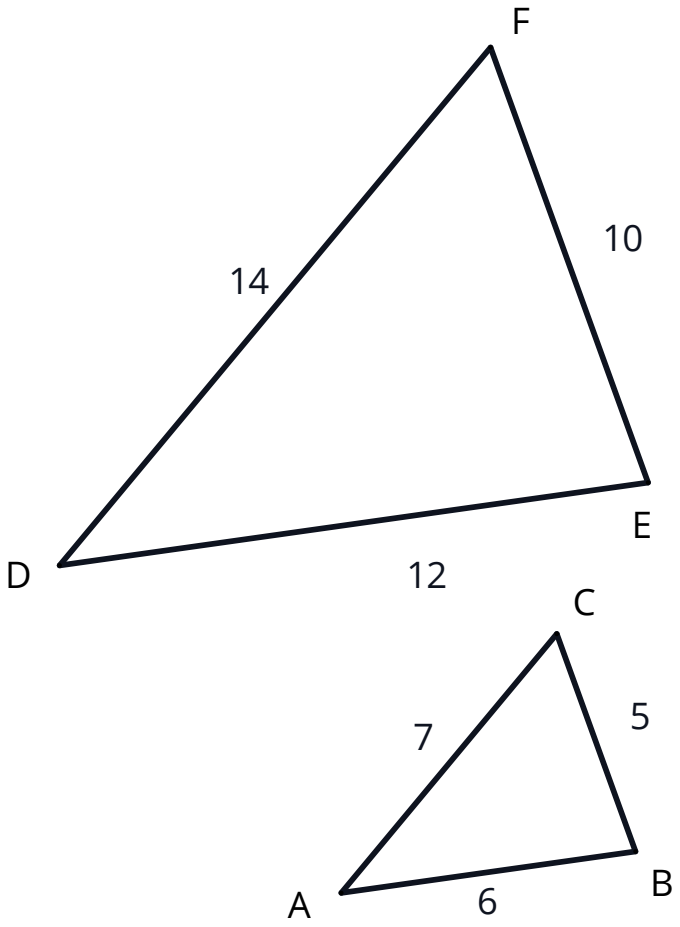
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c



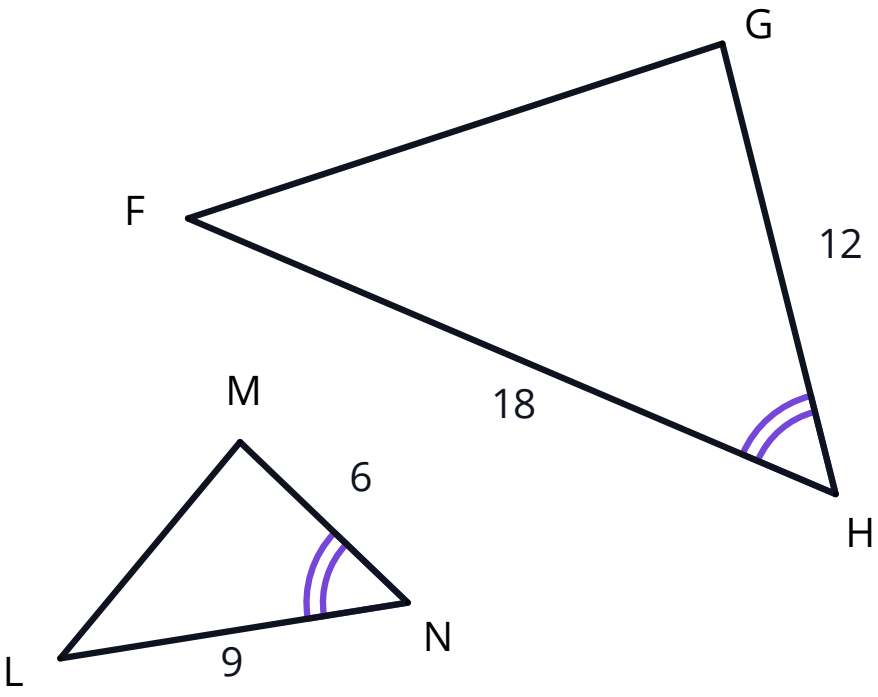
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d



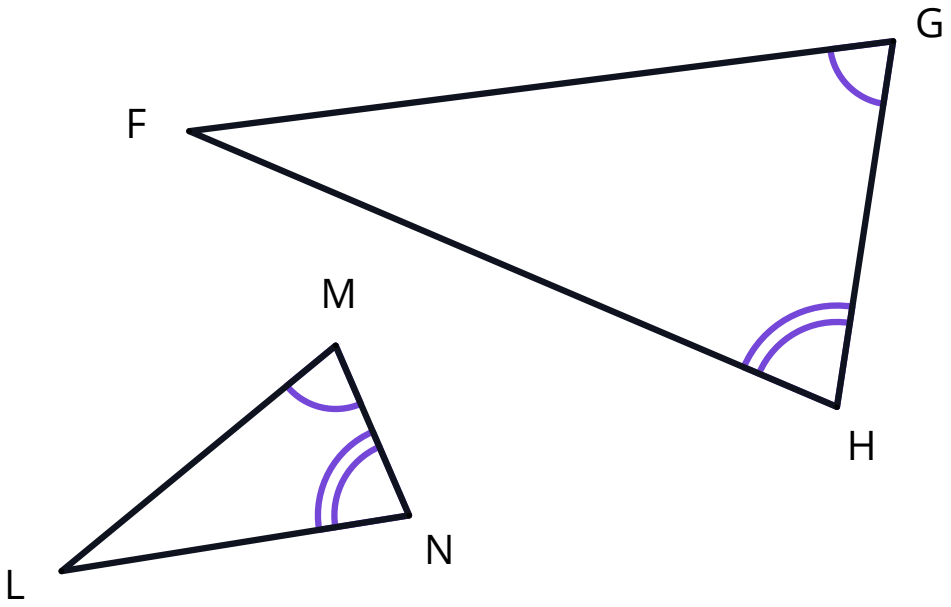
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e



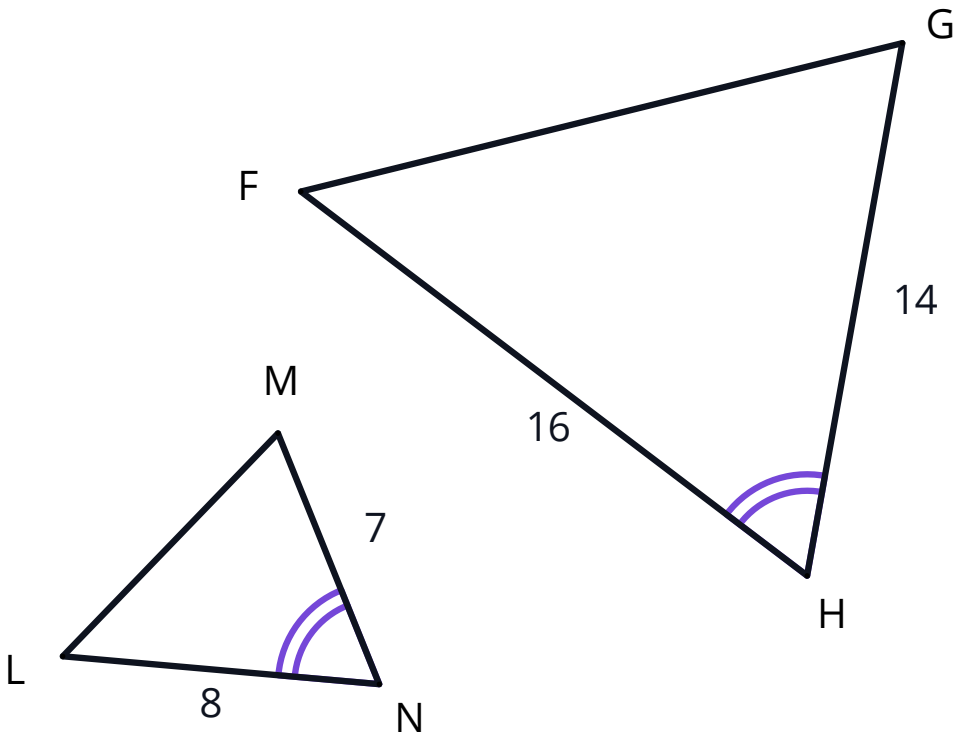
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f



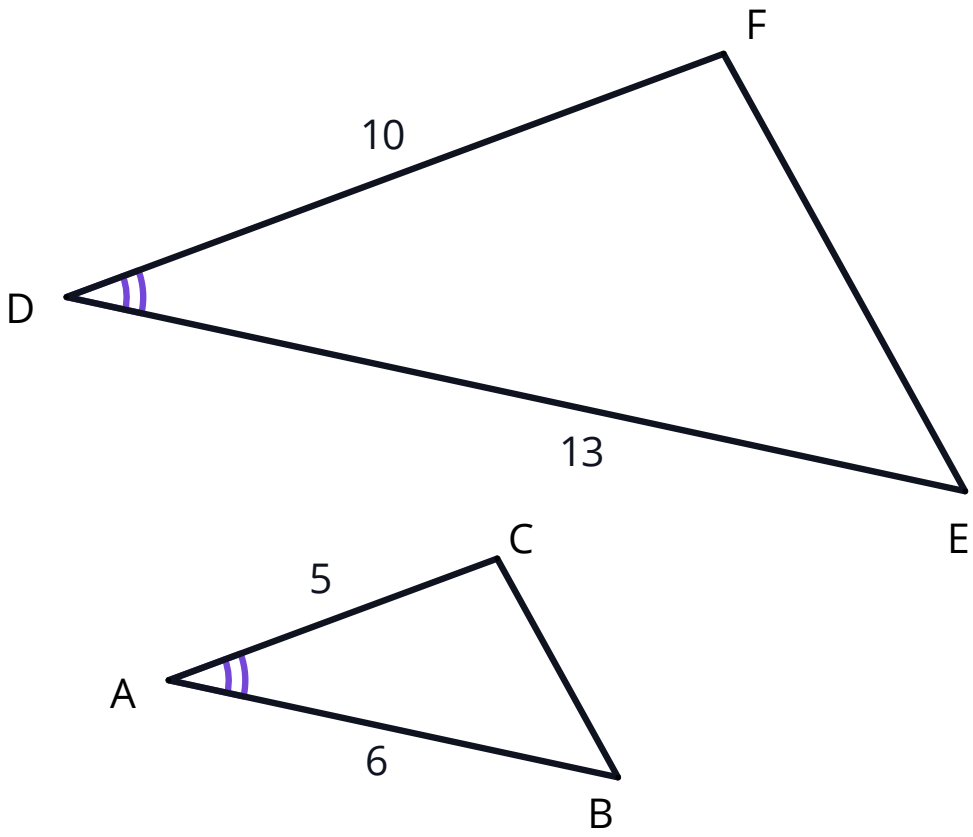
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g



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h



7 Is it true that all equilateral triangles are similar? Justify your reasoning.

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Additional questions

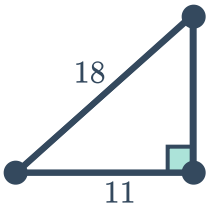
8 For each of the following sets of triangles:

i Identify which two are similar.

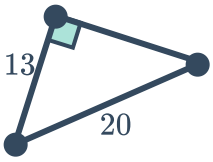
ii Identify the similarity test that the pair satisfy: AA, SSS, SAS, or HL.

a

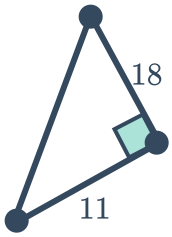
A



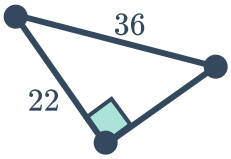
B



C



D

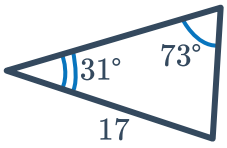


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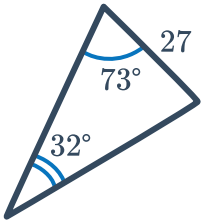
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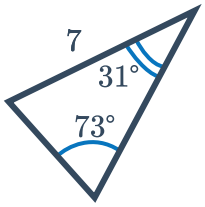
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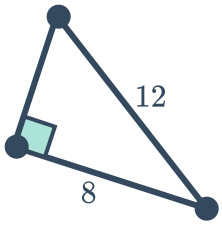


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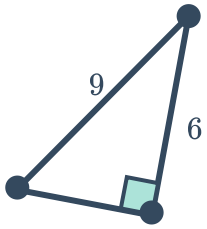


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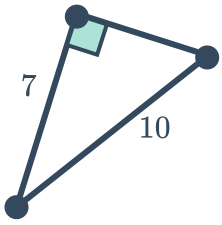
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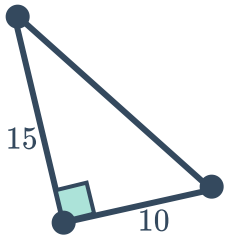
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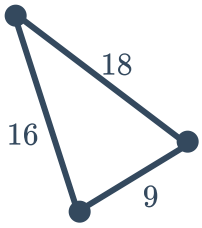


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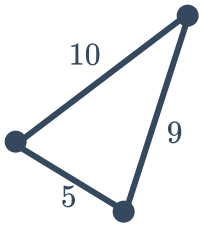


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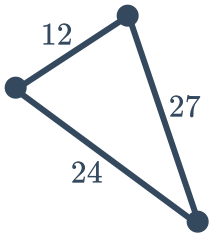
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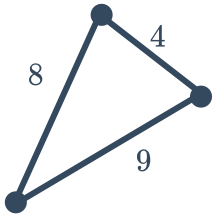
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C



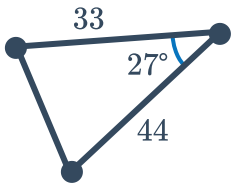
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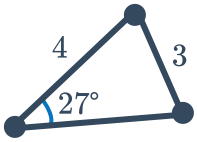
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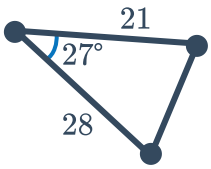
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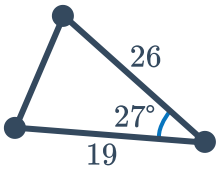
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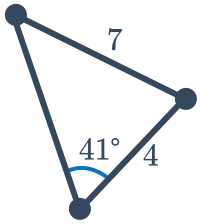


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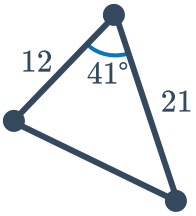


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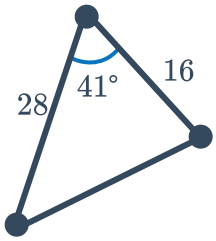
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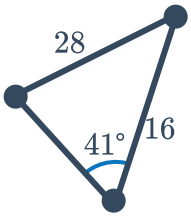
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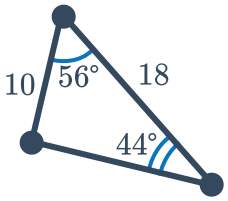


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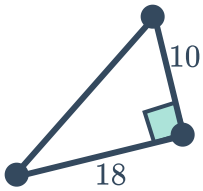


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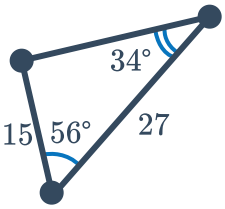
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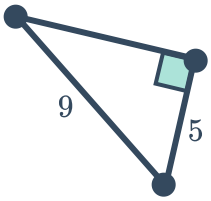
B



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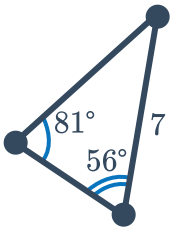


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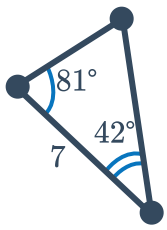


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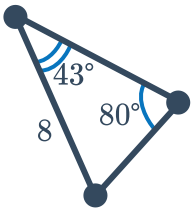
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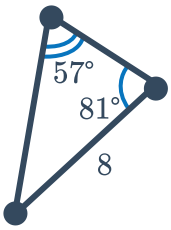
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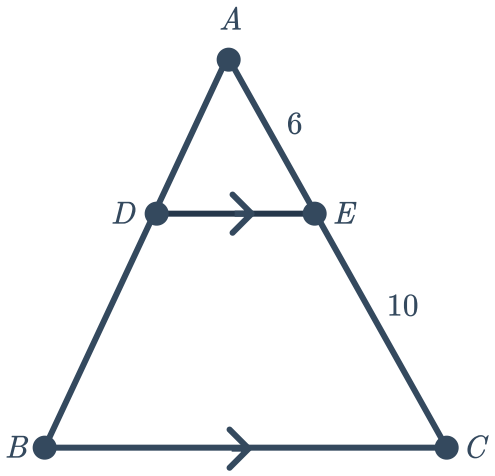
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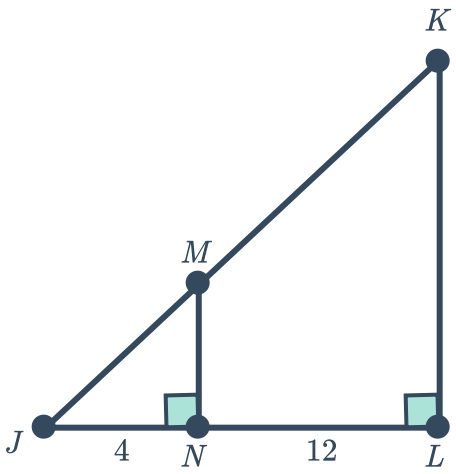
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9 For the following diagrams, identify the two similar triangles and state the similarity test that they satisfy.

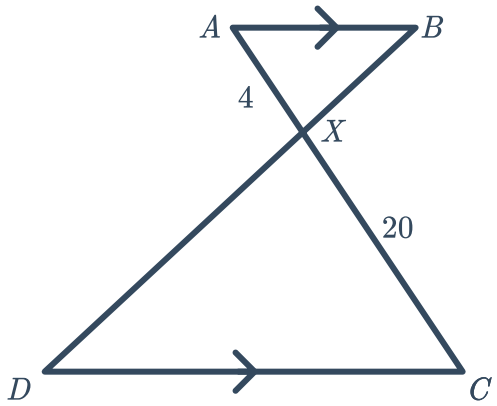
a



b

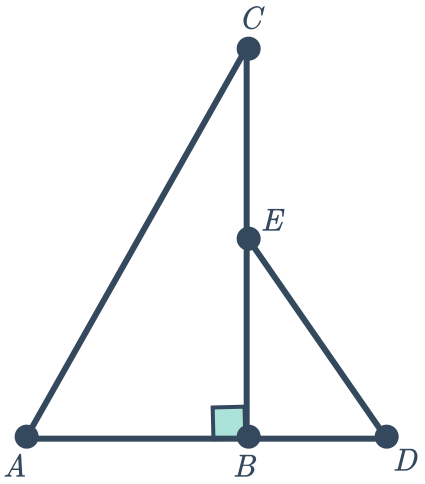


10 Consider the following diagram:
State the similarity test that triangles $\triangle ABX$ and $\triangle CDX$ satisfy.

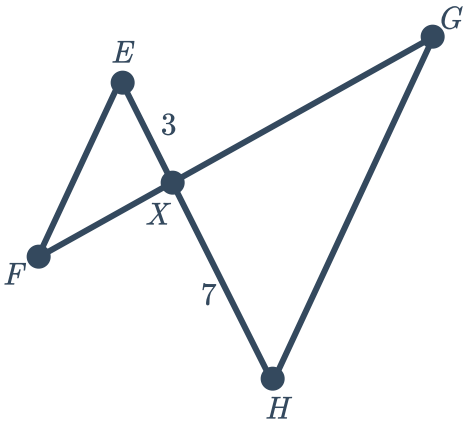


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11 Consider the following diagram, where $\overline{AD}$ is a straight line:
What information is required to know that triangles $\triangle ABC$ and $\triangle DBE$ are in fact similar?



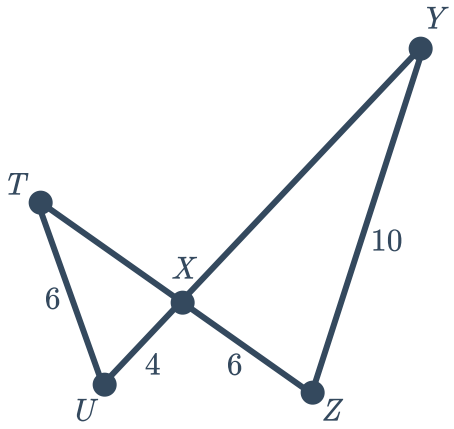
12 Consider the following diagram, where $\overline{EH}$ and $\overline{FG}$ are straight lines:
What information is required to know that triangles $\triangle EFX$ and $\triangle GHX$ are in fact similar?



13 Consider the following diagram, where $\overline{TZ}$ and $\overline{UY}$ are non-parallel straight lines:

a Explain why the triangles $\triangle TUX$ and $\triangle YZX$ are not similar.

b What length does YZ need to be for the two triangles to be similar?



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